

18. ROS2 Rqt Toolbox

Note: The accompanying reference video uses the ROS Humble version, for reference only

- This section's folder contains demonstration GIFs that show the operation process of examples in this section intuitively.
-



3.7.2RQT工具箱.gif



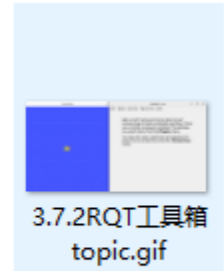
3.7.2RQT工具箱 param.gif



3.7.2RQT工具箱 service.gif



3.7.2RQT工具箱 service-1698722363522-4.gif



3.7.2RQT工具箱 topic.gif

1. Rqt Overview

Rqt is another modular visualization tool provided by ROS. As the name suggests, like Rviz, it is also developed based on the QT visualization tool. Before using it, we need to install it with the following command, and then we can start it using the `rqt` command.

2. Installation

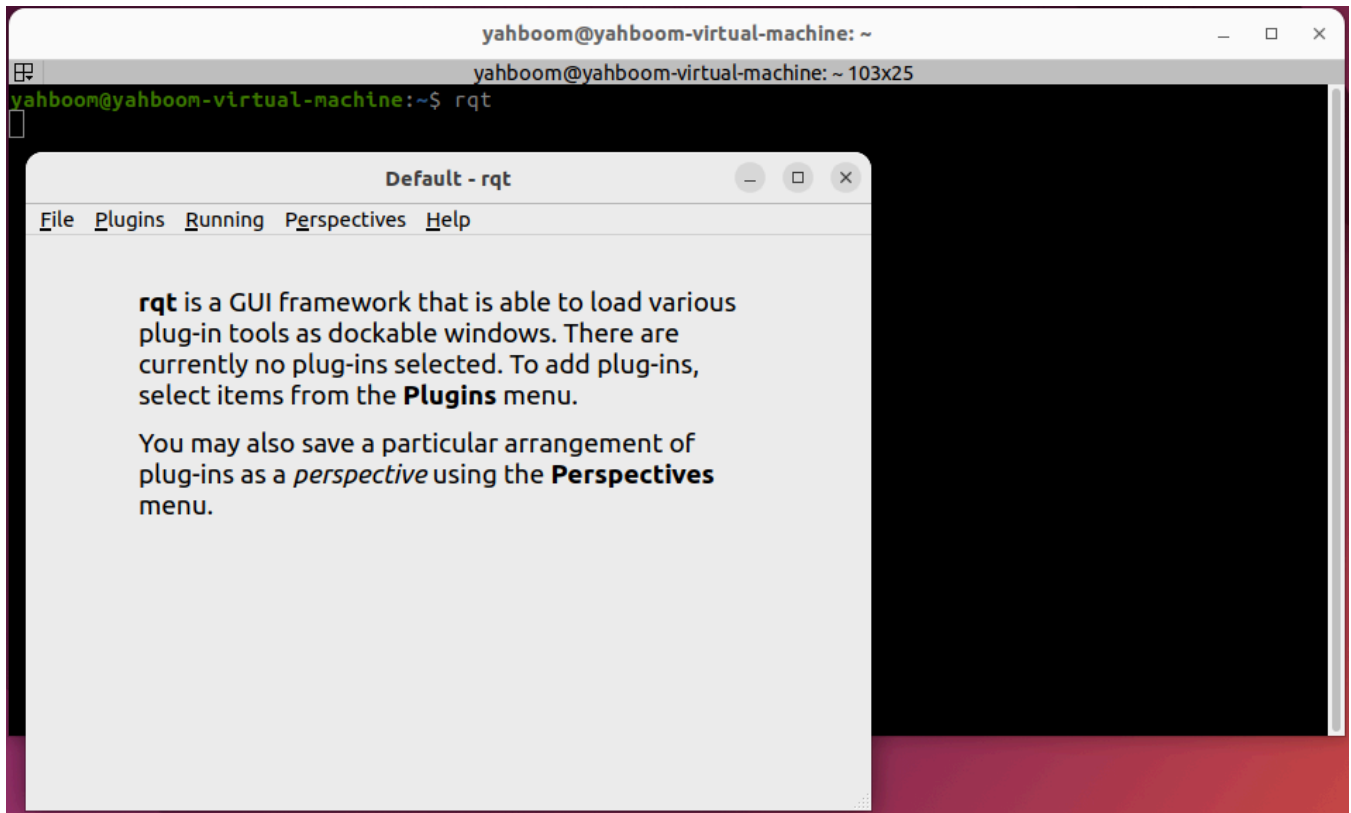
- Generally, if you install the desktop version, rqt toolbox will be installed by default;
- If you didn't install the full version when installing ROS, you can install it as follows:

```
sudo apt install ros-${ROS_DISTRO}-rqt*
```

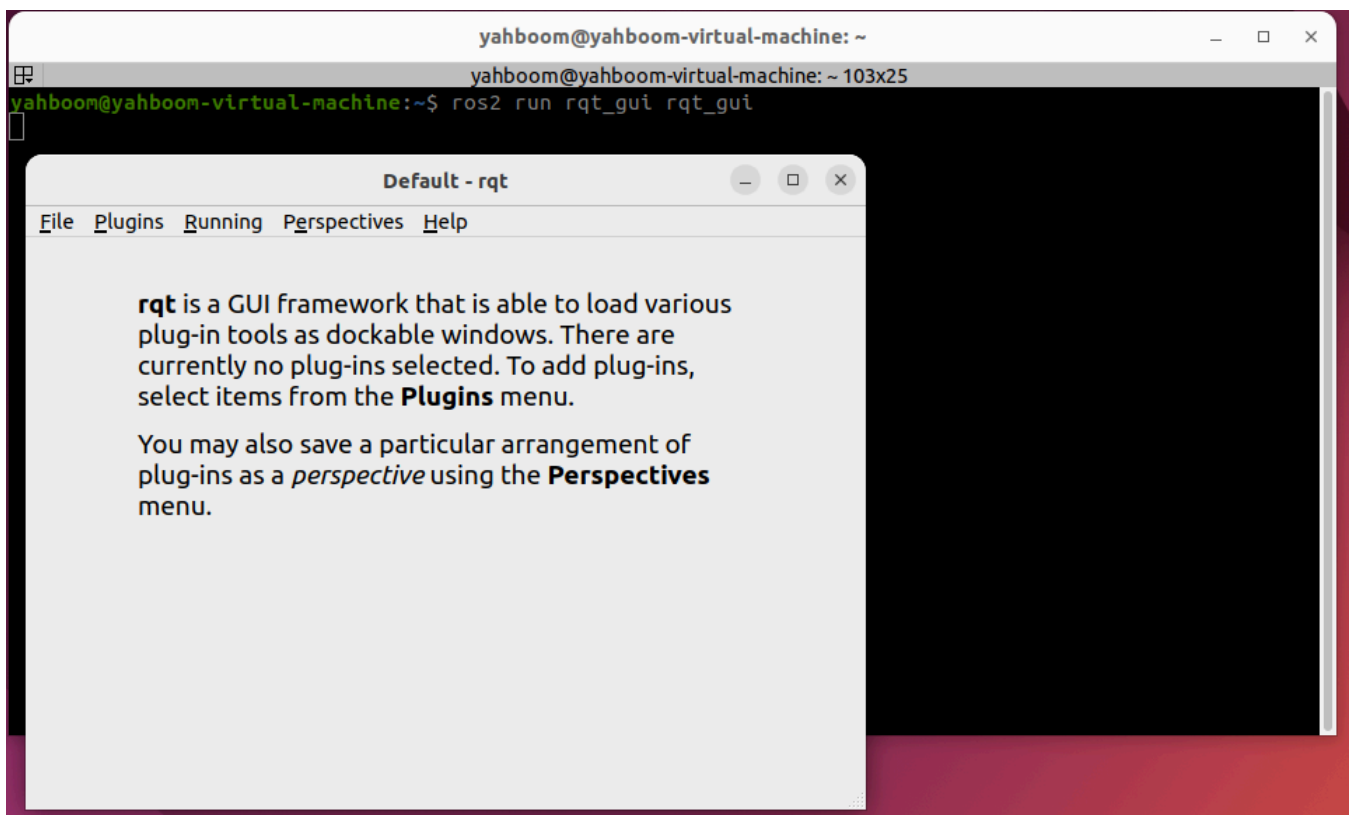
3. Launch

Common `rqt` launch commands are:

- Method 1: `rqt`

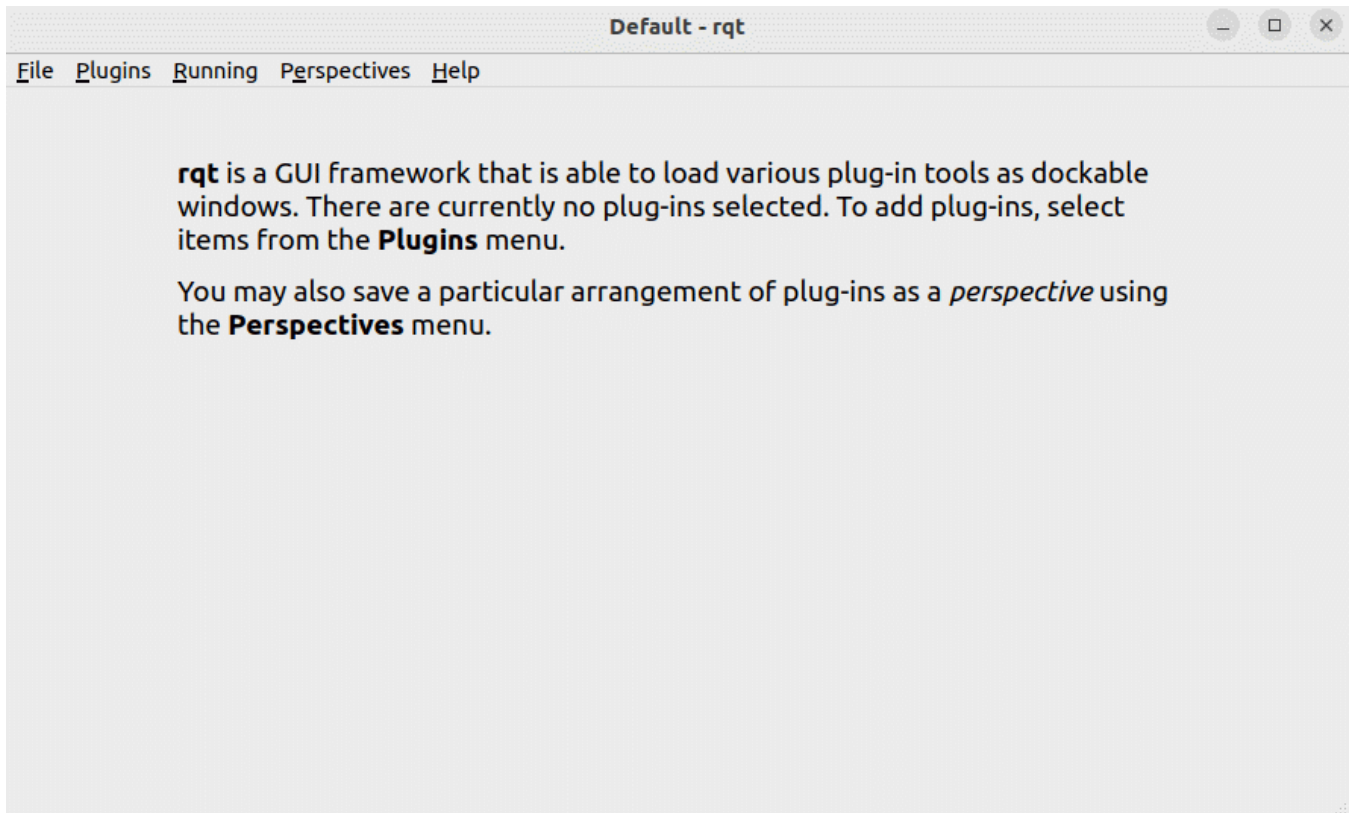


- Method 2: `ros2 run rqt_gui rqt_gui`



4. Plugin Usage

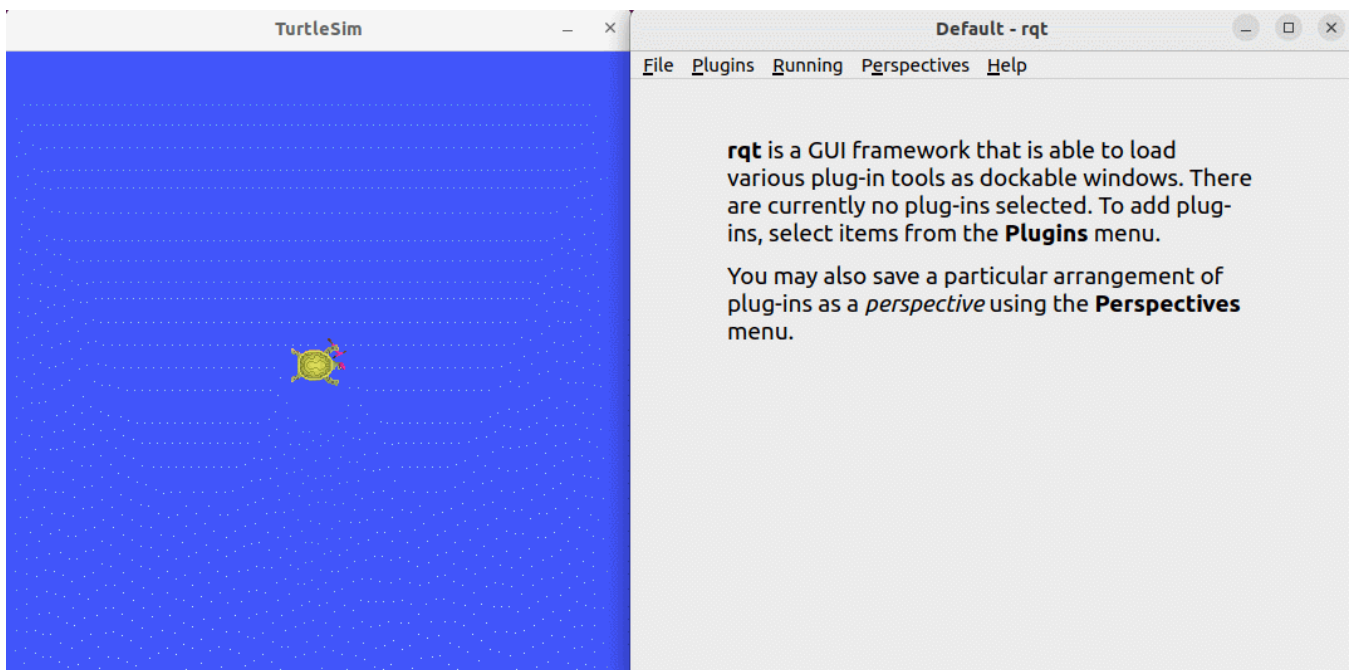
After starting rqt, you can add required plugins through plugins:



The plugins contain topics, services, actions, parameters, logs, and other related plugins. We can choose as needed to conveniently debug ROS2 programs. Usage examples are as follows.

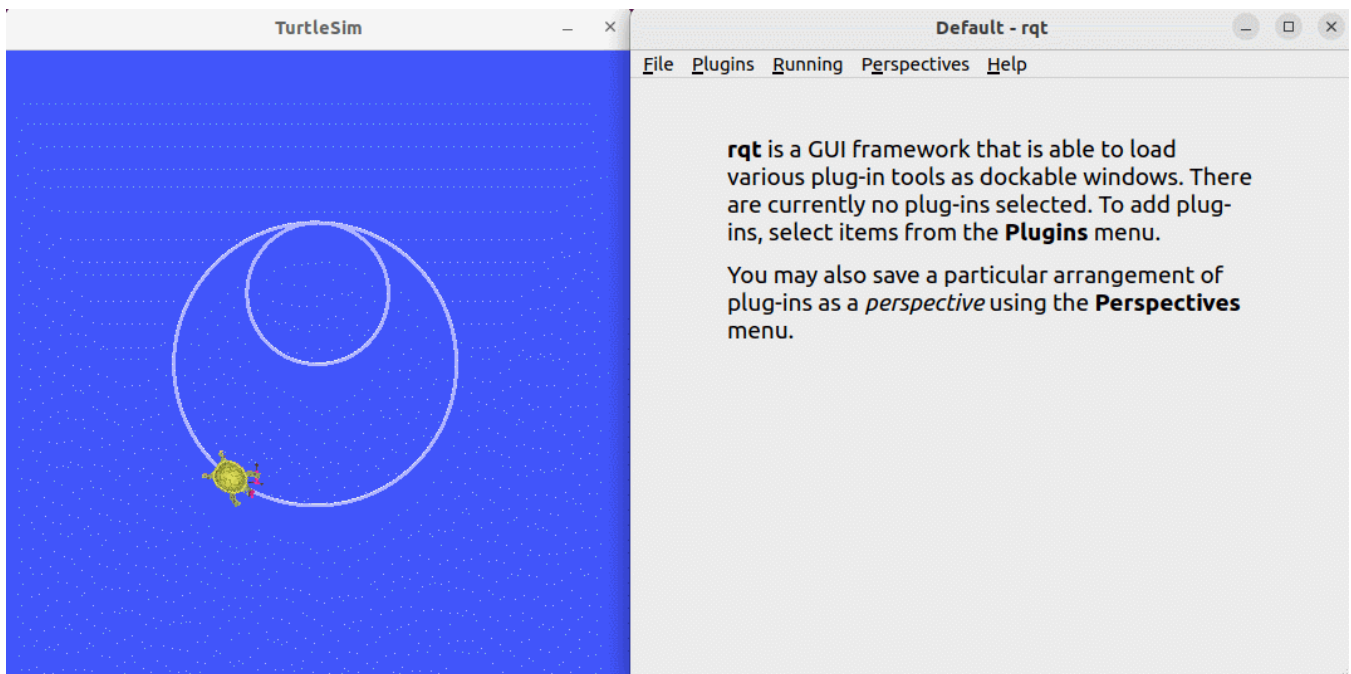
4.1. Topic Plugin

Add topic plugin and send velocity commands to control turtle motion.



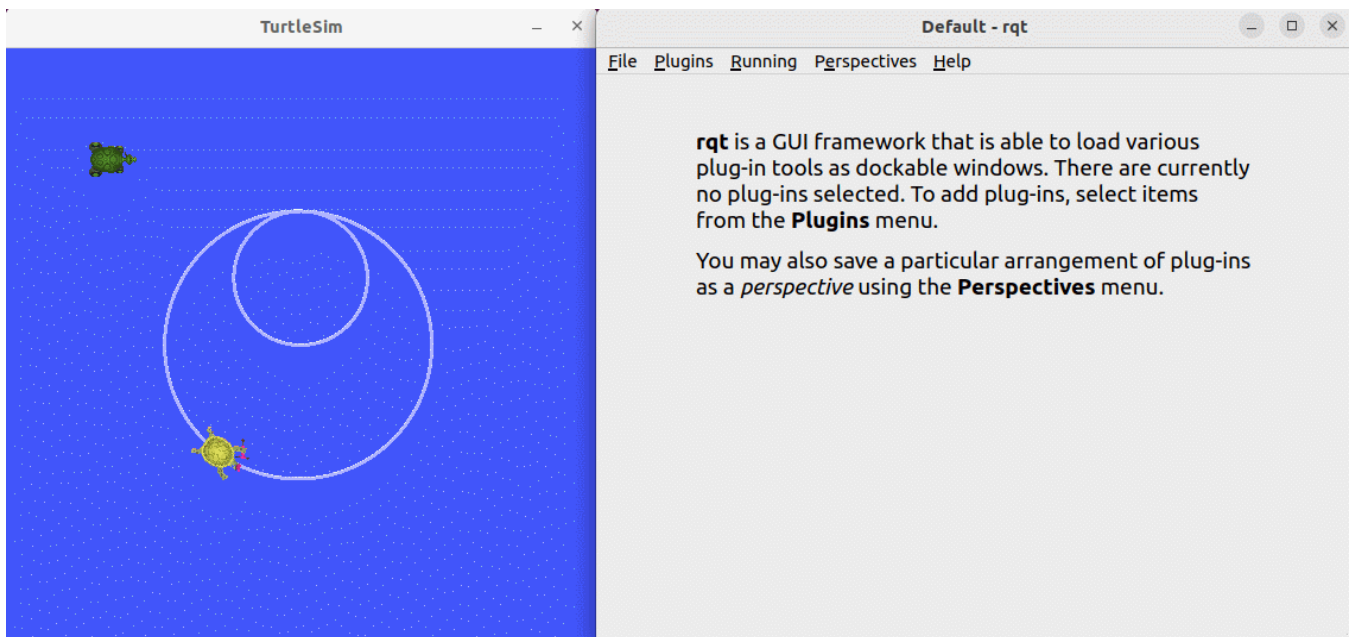
4.2. Service Plugin

Add service plugin and send requests to spawn a turtle at a specified position.



4.3. Parameter Plugin

Dynamically modify turtle window background color through the parameter plugin.



Possible issue: After executing `sudo apt install ros-${ROS_DISTRO}-rqt*`, plugins are not loaded.

Solution: Execute `rqt --force-discover` to force rqt to rescan all installed plugins from the system.